



Session 09: Functions — Push In, Spin Around, Flip Over, and Draw

Phase 2 — Classical Techniques | 90 min

Part A: Numbers Go In, Numbers Come Out

Example 1: Push a Number In, a Number Comes Out

Picture a box called $f(x) = 2x + 3$.

Push in $x = 1$. $\rightarrow$ 5 comes out.

Push in $x = 4$. $\rightarrow$ 11 comes out.

Push in $x = -2$. $\rightarrow$ -1 comes out.

The set of all numbers you can push in is called the **domain**.

The set of all numbers that come out is called the **range**.

For this box, you can push any number in. Any number can come out.

Example 2: Finding Numbers You Can't Push In — Four Rules

Rule 1 — Square Root: $\sqrt{x - 2}$.

The inside of a square root must be 0 or bigger.

Solve $x - 2 \geq 0$. $\rightarrow x \geq 2$.

Numbers smaller than 2 can't go in.

Rule 2 — Denominator: $\frac{1}{x^2-4}$.

If the denominator becomes 0, the box explodes.

Solve $x^2 - 4 = 0$. $\rightarrow x = 2, x = -2$.

You can push in any number except these two.

Rule 3 — Both Mixed: $\frac{\sqrt{x+1}}{x-2}$.

Square root condition: $x + 1 \geq 0 \rightarrow x \geq -1$.

Denominator condition: $x \neq 2$.

Intersect both conditions. $\rightarrow [-1, 2)$ and $(2, \infty)$.

Rule 4 — Logarithm: $\log_3(2x - 5)$.

The expression to the right of log must be positive.

Solve $2x - 5 > 0$. $\rightarrow x > \frac{5}{2}$.

Example 3: What Numbers Come Out

$$f(x) = x^2 + 2.$$

x^2 is always 0 or bigger. Then we add 2.

$\rightarrow$ The numbers that come out are always 2 or bigger.

$$g(x) = \frac{1}{x-1}.$$

As x gets close to 1, the value shoots up to $\pm\infty$.

As x goes to $\pm\infty$, the value gets infinitely close to 0.

$\rightarrow$ Numbers that come out: all real numbers except 0.

$$h(x) = \sqrt{4 - x^2}.$$

x can only go in from -2 to 2 .

At $x = 0$, the peak is 2. At $x = \pm 2$, the floor is 0.

$\rightarrow$ Numbers that come out: 0 to 2.

Part B: Connecting Functions and Swapping Input-Output

Example 4: Connecting Two Functions

Connect $f(x) = 2x + 1$ and $g(x) = x^2$.

Connect g to f — $(f \circ g)(x) = f(g(x))$.

- ① Push x into $g \rightarrow x^2$ comes out.
- ② Push that result into $f \rightarrow 2x^2 + 1$ comes out.

Connect f to g — $(g \circ f)(x) = g(f(x))$.

- ① Push x into $f \rightarrow 2x + 1$ comes out.
- ② Push that result into $g \rightarrow (2x + 1)^2 = 4x^2 + 4x + 1$.

The result changes when you swap the order.

Example 5: Connecting Can Shrink the Domain

$f(x) = \sqrt{x}$, $g(x) = x - 3$.

$f \circ g$: Push x into the inner $g \rightarrow x - 3$.

Push $x - 3$ into the outer $f \rightarrow \sqrt{x - 3}$.

f only accepts 0 or bigger. $x - 3 \geq 0 \rightarrow x \geq 3$.

The domain shrinks to $[3, \infty)$.

$g \circ f$: Push x into the inner $f \rightarrow \sqrt{x}$.

Push $\sqrt{x}$ into the outer $g \rightarrow \sqrt{x} - 3$.

f 's domain is $[0, \infty)$. g accepts anything.

The domain stays $[0, \infty)$.

Example 6: Connecting a Piecewise Function

$$f(x) = \begin{cases} x + 1, & x < 0 \\ x^2, & x \geq 0 \end{cases}$$

Connect $g(x) = 2x - 1$ to it — $(f \circ g)(x)$.

- ① Push x into $g \rightarrow 2x - 1$ comes out.
- ② Check whether this value is less than 0 or 0 or bigger.
- ③ If less than 0, use f 's top rule $(x + 1)$.
If 0 or bigger, use f 's bottom rule (x^2) .

$2x - 1 < 0 \rightarrow x < \frac{1}{2}$ interval: result is $(2x - 1) + 1 = 2x$.

$x \geq \frac{1}{2}$ interval: result is $(2x - 1)^2$.

Example 7: Swapping Input and Output — Inverse Function

$$f(x) = 3x - 6.$$

- ① Write $y = 3x - 6$.
- ② Solve for x : $y + 6 = 3x \rightarrow x = \frac{y+6}{3}$.
- ③ Write x in place of y : $f^{-1}(x) = \frac{x+6}{3}$.

Verify: push $f^{-1}(x)$ into f .

Push $\frac{x+6}{3}$ into $3x - 6 \rightarrow 3 \cdot \frac{x+6}{3} - 6 = x$.

Exactly x comes out. Correct.

Example 8: Inverse of a Rational Function

$$f(x) = \frac{2x+1}{x-3}.$$

- ① $y = \frac{2x+1}{x-3}.$

- ② Multiply both sides by $(x - 3)$: $y(x - 3) = 2x + 1$.
 - ③ Spread out: $yx - 3y = 2x + 1$.
 - ④ Gather x terms on the left: $yx - 2x = 3y + 1$.
 - ⑤ Bundle with x : $x(y - 2) = 3y + 1$.
 - ⑥ Divide by $y - 2$: $x = \frac{3y+1}{y-2}$.
 - ⑦ Write x in place of y : $f^{-1}(x) = \frac{3x+1}{x-2}$.
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Example 9: When You Can't Swap — Snip the Domain

$$f(x) = x^2.$$

Push in $x = 2 \rightarrow 4$. Push in $x = -2 \rightarrow$ also 4.

Different inputs give the same output.

If we swap as-is, one input would produce two outputs. That's not a function.

Keep only the right half: snip the domain to $[0, \infty)$.

$$\rightarrow \text{Inverse: } f^{-1}(x) = \sqrt{x}.$$

Keep only the left half: snip the domain to $(-\infty, 0]$.

$$\rightarrow \text{Inverse: } f^{-1}(x) = -\sqrt{x}.$$

Part C: The Ultimate 7-Step Graph-Drawing Sequence

Get paper and pencil. Follow each step below and draw as you go.

- Step 1 – Find the x -values you can't draw (domain)
- Step 2 – If symmetric, draw only half
- Step 3 – Mark where it meets the axes (intercepts)
- Step 4 – Draw lines it approaches but never touches (asymptotes)
- Step 5 – Judge the sign – above or below the x -axis
- Step 6 – Connect the dots with a smooth curve
- Step 7 – Steps, sawteeth, or pieces: draw each interval separately

Example 10: Polynomial — Draw Using Symmetry and End Behavior

$$f(x) = x^3 - 4x.$$

Step 1 — Domain: A polynomial can be drawn for all x .

Step 2 — Symmetry: Compute $f(-x)$.

$$(-x)^3 - 4(-x) = -x^3 + 4x = -(x^3 - 4x) = -f(x).$$

→ Origin symmetry. Draw only the right side, then spin it around and copy.

Step 3 — Intercepts: Find where it meets the x -axis.

$$x^3 - 4x = 0 \rightarrow x(x^2 - 4) = 0 \rightarrow x(x - 2)(x + 2) = 0.$$

It crosses the x -axis at $x = -2, 0, 2$.

Plug in $x = 0 \rightarrow f(0) = 0$. It passes through the origin.

Step 5 — Sign: The roots $-2, 0, 2$ split the line into four intervals.

$x < -2$: all three factors negative → product negative → below axis.

$-2 < x < 0$: two negative, one positive → product positive → above axis.

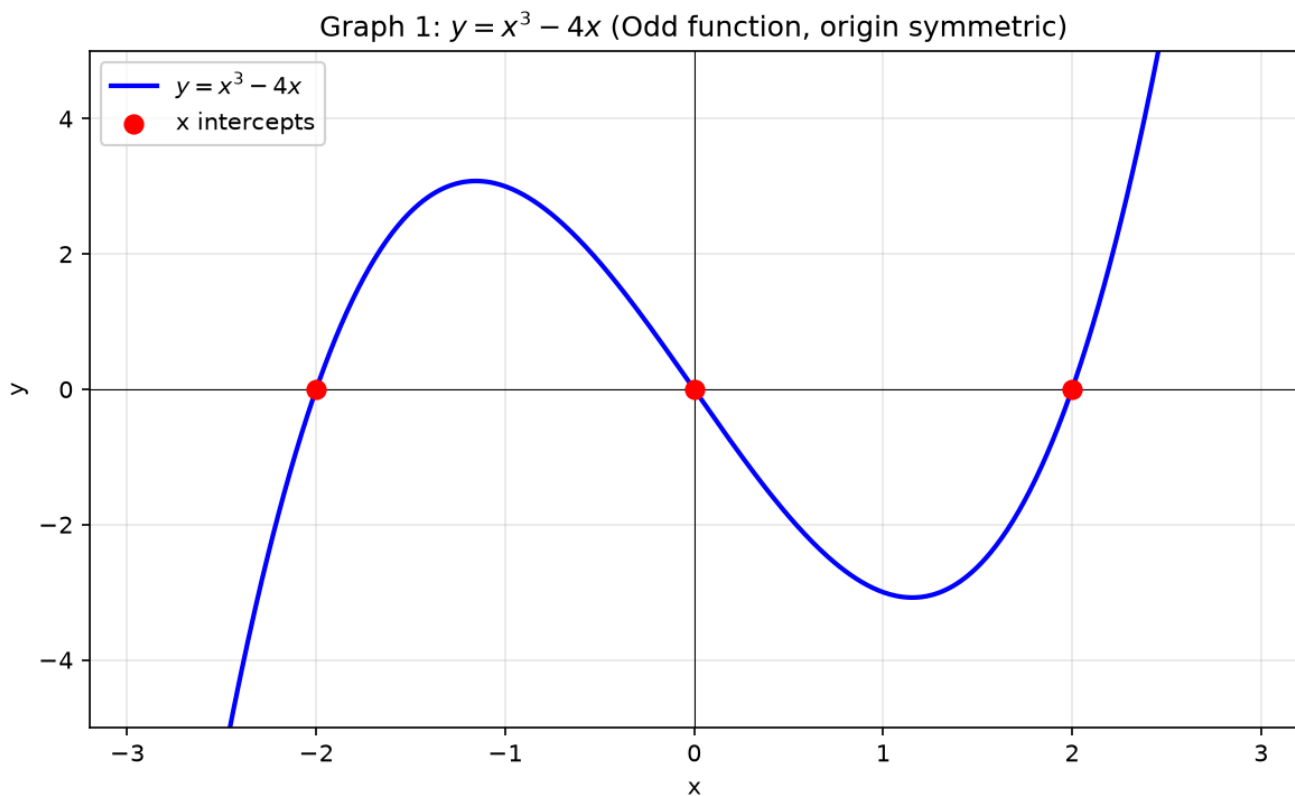
$0 < x < 2$: one negative, two positive → product negative → below axis.

$x > 2$: all three positive → product positive → above axis.

End behavior: $x \rightarrow \infty \rightarrow x^3$ dominates → $+\infty$.

$x \rightarrow -\infty \rightarrow x^3$ dominates → $-\infty$.

Step 6 — Connect: Start from bottom left → cross axis upward at $x = -2$ → come up then down, cross origin → cross axis upward at $x = 2$ → head to top right.



Example 11: Rational Function — Tear and Cancel First

$$f(x) = \frac{x^2 - x - 2}{x^2 - 4}.$$

Step 0 — Tear: Numerator $(x - 2)(x + 1)$, denominator $(x - 2)(x + 2)$.

$(x - 2)$ appears in both numerator and denominator. → It cancels.

$$f(x) = \frac{x+1}{x+2}, \text{ but } x \neq 2.$$

$x = 2$ made the original denominator 0. It canceled but still can't go in.

→ Punch an **empty circle (hole)** at $(2, \frac{3}{4})$.

Step 1 — Domain: $x \neq -2$ (denominator 0), $x \neq 2$ (hole).

Step 4 — Asymptotes:

Vertical: $x = -2$ where denominator is 0.

Horizontal: as $x \rightarrow \infty$, $\frac{x+1}{x+2} \rightarrow 1$. → $y = 1$.

Step 3 — Intercepts:

y -intercept: plug in $x = 0 \rightarrow f(0) = \frac{1}{2}$.

x -intercept: numerator=0 $\rightarrow x = -1$.

Step 5 — Sign near asymptotes:

$x \rightarrow -2^+$: numerator negative, denominator tiny positive $\rightarrow -\infty$.

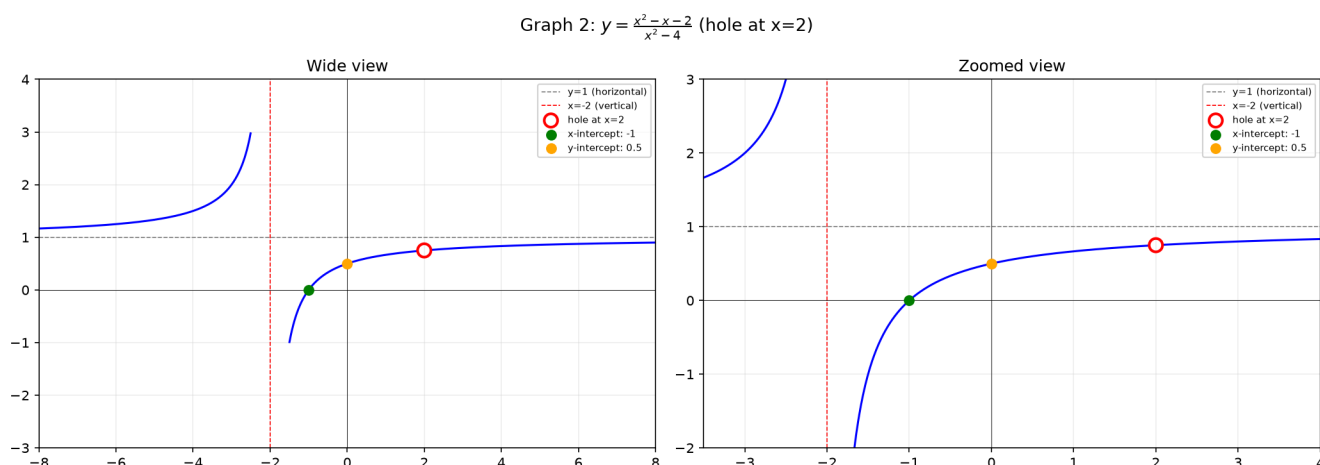
$x \rightarrow -2^-$: numerator negative, denominator tiny negative $\rightarrow +\infty$.

Step 6 — Connect:

Left of $x = -2$: starts at $+\infty$, passes x -intercept -1 , approaches $y = 1$ from below.

Right of $x = -2$: starts at $-\infty$, passes y -intercept $\frac{1}{2}$, approaches $y = 1$ from below.

Empty circle at $x = 2$.



Example 12: Slant Asymptote — Divide First

$$f(x) = \frac{x^2 + 2x}{x - 1}.$$

Divide: $x^2 + 2x$ divided by $x - 1$.

Quotient $x + 3$, remainder 3.

$$\rightarrow f(x) = x + 3 + \frac{3}{x - 1}.$$

Vertical asymptote: denominator 0 $\rightarrow x = 1$.

Slant asymptote: as $x \rightarrow \pm\infty$, $\frac{3}{x - 1} \rightarrow 0$.

$\rightarrow$ The graph hugs the line $y = x + 3$.

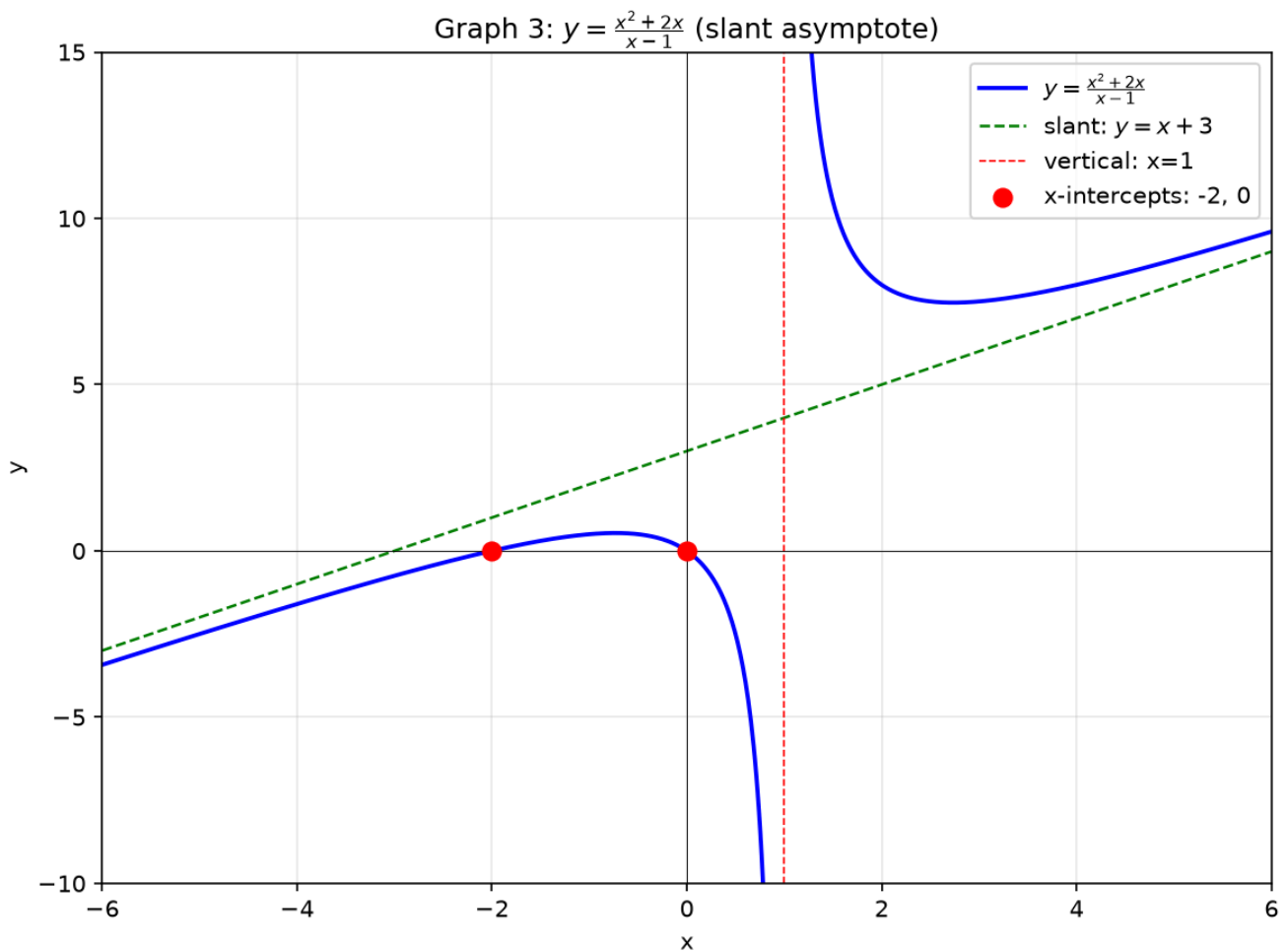
Intercepts: $x = 0 \rightarrow f(0) = 0$ (origin). $f(x) = 0 \rightarrow x(x + 2) = 0 \rightarrow x = 0, -2$.

Near asymptote: $x \rightarrow 1^+ \rightarrow \frac{3}{x-1} \rightarrow +\infty \rightarrow +\infty$.

$x \rightarrow 1^- \rightarrow \frac{3}{x-1} \rightarrow -\infty \rightarrow -\infty$.

Drawing order:

- ① Draw the slant line $y = x + 3$ as a dashed line first.
- ② Draw the vertical dashed line at $x = 1$.
- ③ Mark $(-2, 0)$ and $(0, 0)$.
- ④ Draw the curve hugging the asymptotes.



Example 13: Hyperbola — The $\frac{ax+b}{cx+d}$ Form

$$f(x) = \frac{2x+1}{x-1}.$$

Vertical asymptote: denominator $0 \rightarrow x = 1$.

Horizontal asymptote: $x \rightarrow \infty, \frac{2x}{x} \rightarrow 2 \rightarrow y = 2$.

Intercepts:

y -intercept: $x = 0 \rightarrow y = -1$.

x -intercept: $2x + 1 = 0 \rightarrow x = -\frac{1}{2}$.

Center: the intersection of the two asymptotes $(1, 2)$.

The graph is a hyperbola, symmetric on opposite sides of this point.

Approach directions:

$x \rightarrow 1^+$: $(+)/(+ \text{ small}) \rightarrow +\infty$.

$x \rightarrow 1^-$: $(+)/(- \text{ small}) \rightarrow -\infty$.

$x \rightarrow \infty$: $2 + \frac{3}{x-1} \rightarrow$ approaches $y = 2$ from above.

$x \rightarrow -\infty$: approaches $y = 2$ from below.

Example 14: Radical Function — A Half-Graph

$$f(x) = \sqrt{x-1} + 2.$$

Starting point: only at $x = 1$ does the inside of the root become 0.

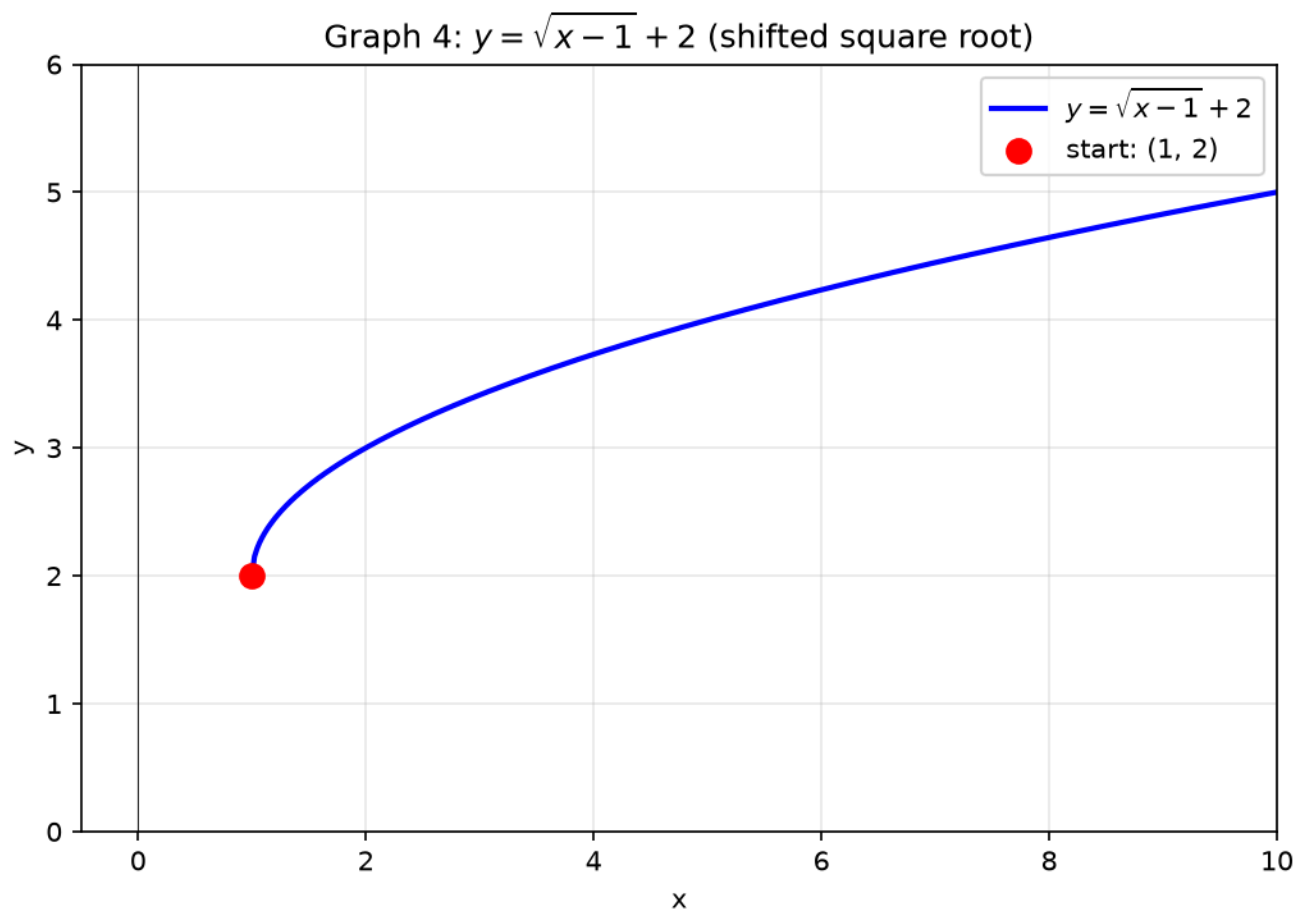
$f(1) = 0 + 2 = 2$. Mark the point $(1, 2)$.

For $x < 1$, the inside is negative $\rightarrow$ no graph.

Growth: as x gets bigger, the root value grows slowly.

The whole graph spreads to the upper right.

Shape: $y = \sqrt{x}$ shifted right by 1 and up by 2.



Part D: Steps, Sawteeth, and Signs — Special Functions

Example 15: The Floor Function $\lfloor x \rfloor$ — Building Stairs

$\lfloor x \rfloor$ is the greatest integer not exceeding x .

Let's pin down some values:

$$\lfloor 0 \rfloor = 0, \lfloor 0.3 \rfloor = 0, \lfloor 0.999 \rfloor = 0.$$

$$\lfloor 1 \rfloor = 1, \lfloor 1.7 \rfloor = 1.$$

$$[-0.3] = -1, [-1] = -1, [-1.2] = -2.$$

Pattern: the interval $[0, 1)$ is all at height 0.

$[1, 2)$ is all at height 1.

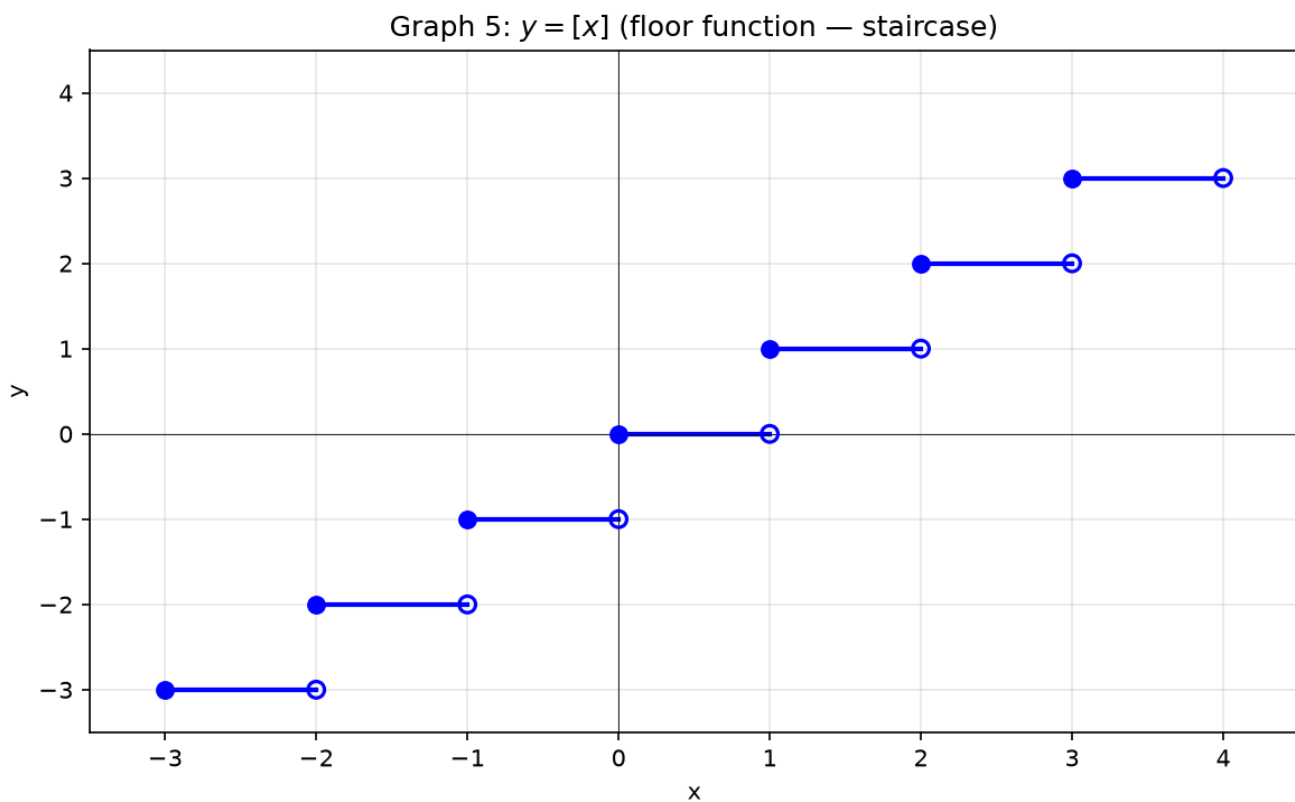
$[-1, 0)$ is all at height -1 .

How to draw:

On $[0, 1)$, draw a horizontal segment at height 0. Right endpoint $(1, 0)$ is an empty circle.

On $[1, 2)$, draw a horizontal segment at height 1. Left endpoint $(1, 1)$ is a filled circle.

Repeat left and right.



$[2x]$: the stair width is halved. $[0, 0.5)$ is 0, $[0.5, 1)$ is 1.

$[x] + [-x]$: 0 for integers, -1 otherwise.

Example 16: The Fractional Part $\{x\}$ — Repeating Sawteeth

$\{x\} = x - [x]$. Strip away the integer part, leaving only the decimal.

$\{3.7\} = 0.7$, $\{5.0\} = 0$, $\{-1.2\} = -1.2 - (-2) = 0.8$.

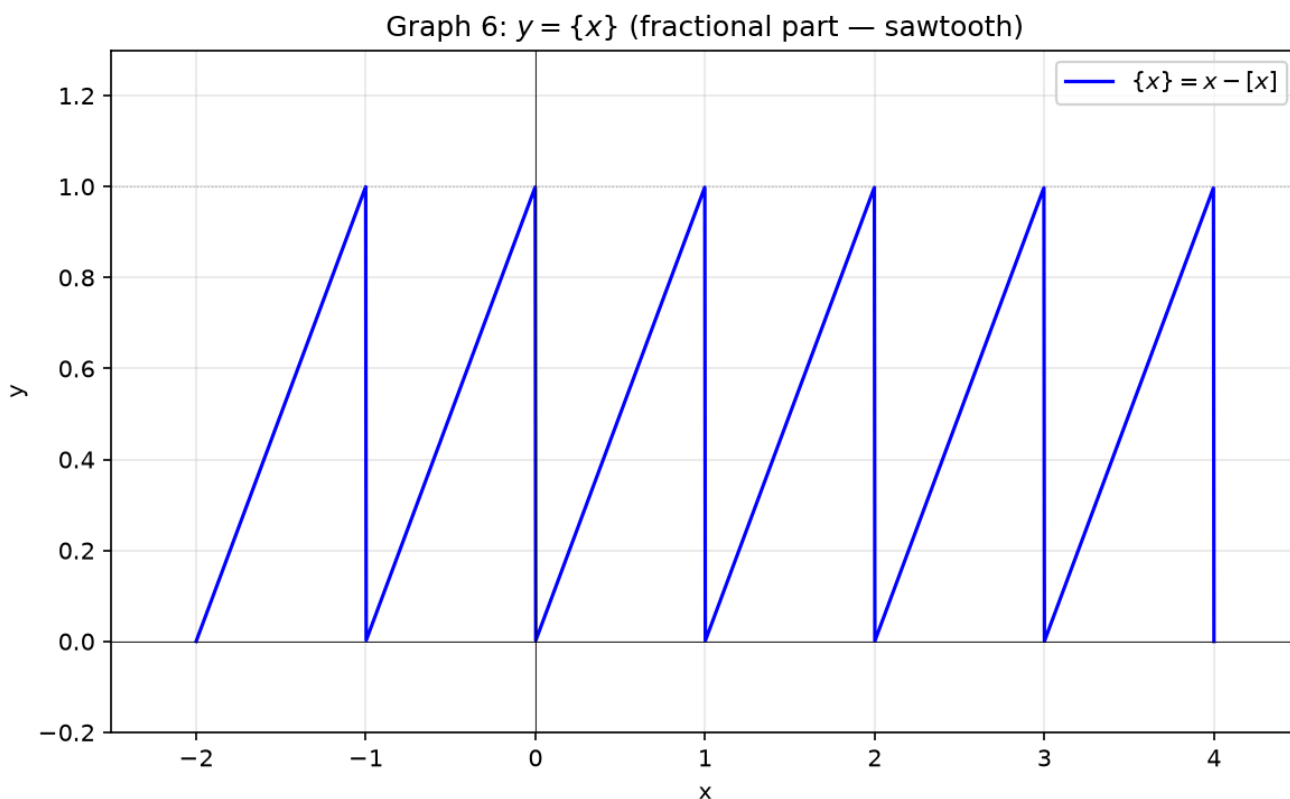
How to draw:

$[0, 1)$ interval: $\{x\} = x - 0 = x$. A diagonal from the origin to just before $(1, 1)$.

$[1, 2)$ interval: $\{x\} = x - 1$. A diagonal from $(1, 0)$ to just before $(2, 1)$.

$[-1, 0)$ interval: $\{x\} = x + 1$. A diagonal from $(-1, 0)$ to just before $(0, 1)$.

The sawtooth repeats infinitely with height in $[0, 1)$. 1 is never reached.



Example 17: Ceiling Function and Sign Function

$\lceil x \rceil$: the smallest integer not less than x .

$$\lceil 3.2 \rceil = 4, \lceil -1.2 \rceil = -1.$$

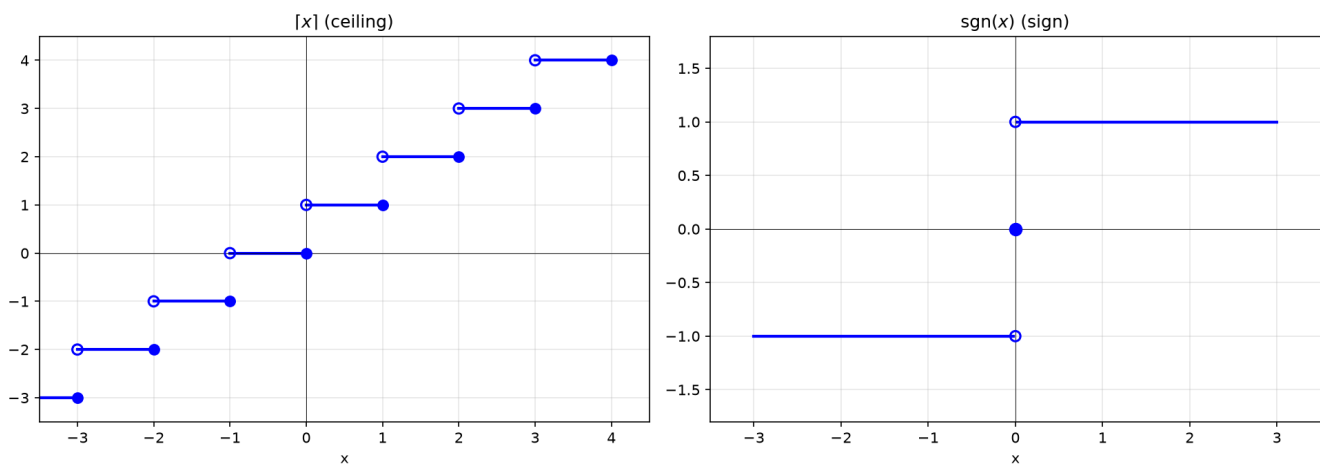
On $(0, 1]$ it's 1, on $(1, 2]$ it's 2. Like the floor function but shifted right.

$\text{sgn}(x)$: keeps only the sign.

If $x < 0$, -1 . If $x = 0$, 0 . If $x > 0$, 1 .

The graph is three horizontal segments at heights -1 , 0 , 1 .

Graph 7: Ceiling & Sign Functions



Part E: Cooking with Graphs — Move, Flip, and Stretch

Example 18: Move the Whole Graph

Base ingredient: $f(x) = |x|$. A V-shape, vertex at the origin.

Right by 3: $f(x - 3) = |x - 3|$.

Push $x - 3$ into the x slot. $\rightarrow$ the vertex moves to $(3, 0)$.

Up by 2: $f(x) + 2 = |x| + 2$.

Add 2 to everything. $\rightarrow$ the vertex moves to $(0, 2)$.

Left 1, down 4: $f(x + 1) - 4 = |x + 1| - 4$.

$\rightarrow$ the vertex moves to $(-1, -4)$.

By hand: draw the original graph, move the vertex, redraw.

Example 19: Flip and Reflect

Ingredient: $f(x) = \sqrt{x}$. Only exists for $x \geq 0$, spreads to the upper right.

Flip over the x -axis: $y = -\sqrt{x}$.

$(4, 2)$ drops to $(4, -2)$. The graph points downward.

Flip over the y -axis: $y = \sqrt{-x}$.

Only exists for $x \leq 0$. Mirror image of the right-side graph.

Flip over the origin: $y = -\sqrt{-x}$.

Flipped twice. Only exists in quadrant 3.

Example 20: Fold the Part Below the Axis Upward

$y = |f(x)|$ — **Fold the part below the x -axis upward.**

Ingredient: $f(x) = x^2 - 1$.

A parabola dipping down to -1 at $x = 0$.

$|f(x)| = |x^2 - 1|$:

The interval $-1 < x < 1$ was originally below the axis.

Fold this part upward across the x -axis.

$\rightarrow$ Between $(-1, 0)$ and $(1, 0)$ becomes an upward bump.

$y = f(|x|)$ — **Copy the right side onto the left.**

Ingredient: $f(x) = x^2 - 2x$.

$$f(|x|) = |x|^2 - 2|x| = x^2 - 2|x|.$$

For $x \geq 0$, it's the original graph.

For $x < 0$, reflect the right-side graph across the y -axis and stick it on.

→ A W-shape.

Example 21: Stretch and Shrink

Ingredient: $f(x) = \sin x$. Period 2π , height 1.

Vertical stretch: $2 \sin x \rightarrow$ height doubles.

Vertical shrink: $\frac{1}{2} \sin x \rightarrow$ height halves.

Horizontal stretch: $\sin\left(\frac{x}{2}\right) \rightarrow$ period doubles to 4π .

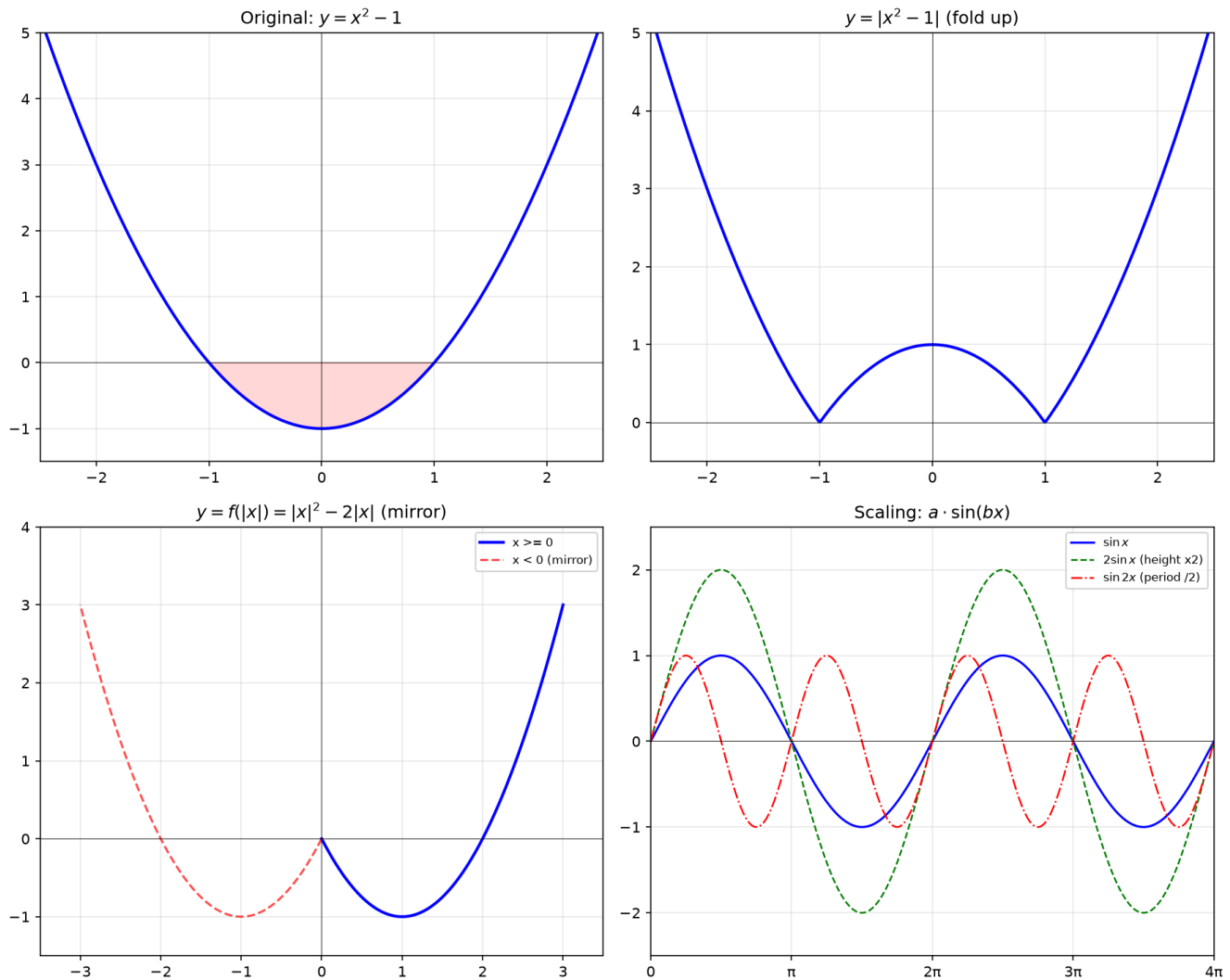
Horizontal shrink: $\sin(2x) \rightarrow$ period halves to π .

Rule: in $a \cdot f(bx)$,

a scales the height by a . b scales the horizontal speed by b .

$b > 1$ squeezes it, $b < 1$ stretches it.

Graph 8: Absolute Value & Scaling Transformations



Example 22: Piecewise Function — Draw Each Piece and Stick Together

$$f(x) = \begin{cases} x + 2, & x \leq 0 \\ 4 - x^2, & 0 < x \leq 2 \\ \frac{1}{x-2}, & x > 2 \end{cases}$$

Check the boundaries first:

$x = 0$: first piece gives 2. Second piece gives 4. → **Discontinuity!**

Mark a filled dot at $(0, 2)$ and an empty dot at $(0, 4)$.

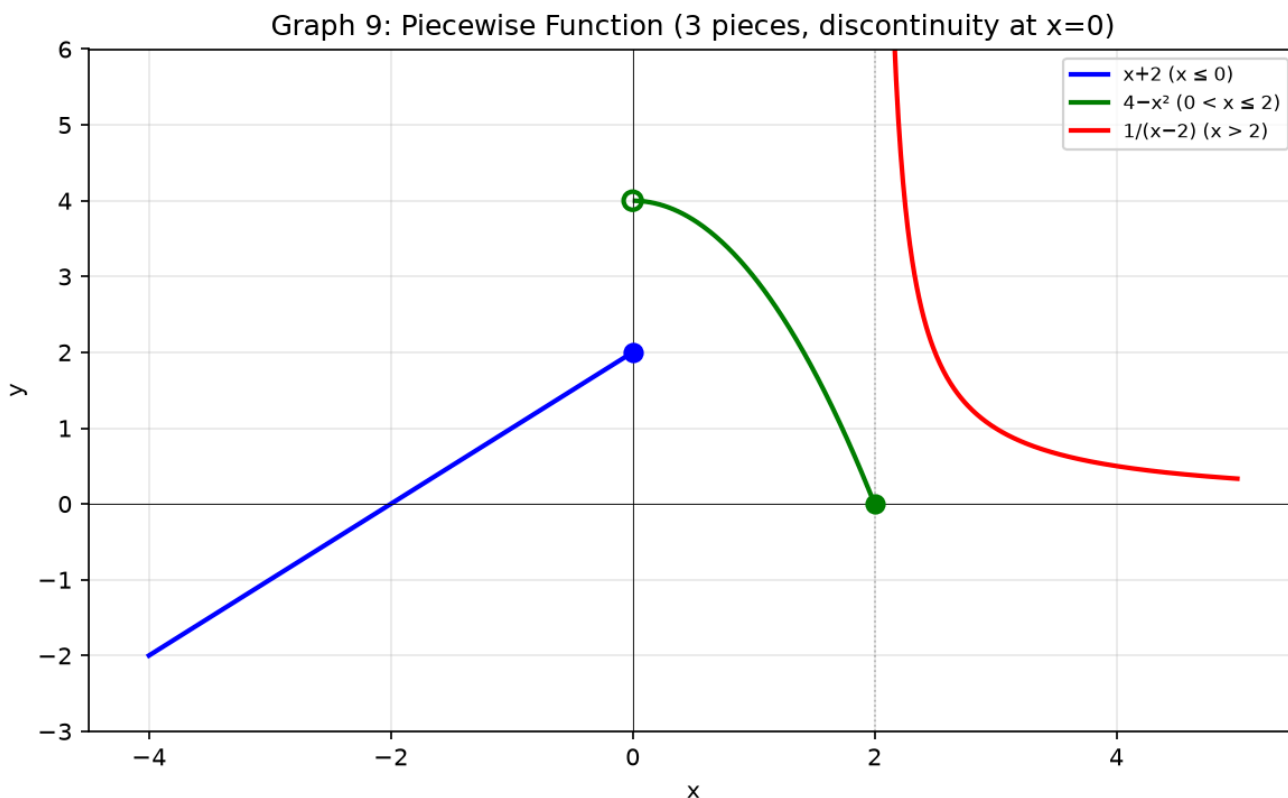
$x = 2$: second piece ends at 0. Third piece has denominator 0 at $x = 2$.
 $\rightarrow x = 2$ is a vertical asymptote.

Draw piece by piece:

Piece 1: $x \leq 0$, line $y = x + 2$. Ends at $(0, 2)$.

Piece 2: $0 < x \leq 2$, downward opening parabola. Starts from empty dot $(0, 4)$, ends at $(2, 0)$.

Piece 3: $x > 2$, $y = \frac{1}{x-2}$. Wall at $x = 2$, approaches 0 going right.



Example 23: Mixing $[x]$ and $\{x\}$

$$f(x) = x + [x].$$

Write x as $n + \delta$ ($0 \leq \delta < 1$, n integer).

$$f(x) = n + \delta + n = 2n + \delta.$$

$[0, 1)$: $f(x) = 0 + \delta = \delta$. Diagonal from $(0, 0)$ to just before $(1, 1)$.

$[1, 2)$: $f(x) = 2 + \delta$. From $(1, 2)$ to just before $(2, 3)$.

Jumps by 1 at every integer.

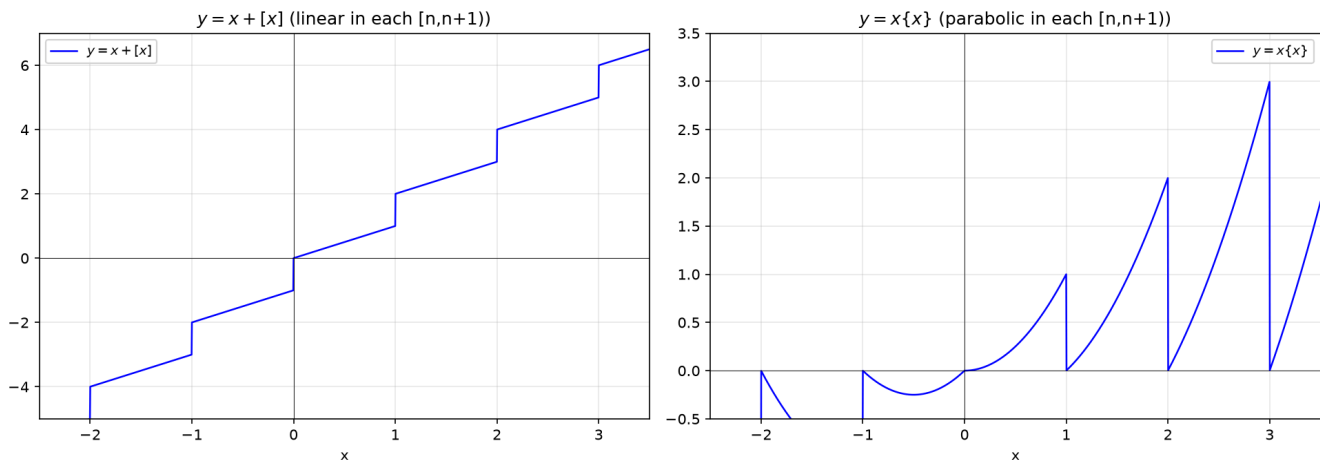
$$f(x) = x\{x\} = x(x - [x]).$$

$[0, 1)$: $x \cdot x = x^2$. One piece of a parabola.

$[1, 2)$: $x(x - 1) = x^2 - x$. Another piece of a parabola.

Drops to 0 at every integer.

Graph 10: Mixed Floor/Fractional Part Functions



Example 24: Putting It All Together — Absolute Value + Rational

$$f(x) = \frac{|x-1|}{x^2-1}.$$

Domain: denominator $x^2 - 1 = (x - 1)(x + 1) \rightarrow x \neq 1, -1$.

Strip the absolute value:

When $x \geq 1$, $|x - 1| = x - 1 \rightarrow f(x) = \frac{x-1}{(x-1)(x+1)} = \frac{1}{x+1} \ (x \neq 1)$.

When $x < 1$, $|x - 1| = -(x - 1) \rightarrow f(x) = \frac{-1}{x+1} \ (x \neq -1)$.

Hole: $x = 1$ canceled but was originally excluded.

Plug $x = 1$ into the $x \geq 1$ formula $\rightarrow \frac{1}{2}$.

$\rightarrow$ Empty circle at $(1, \frac{1}{2})$.

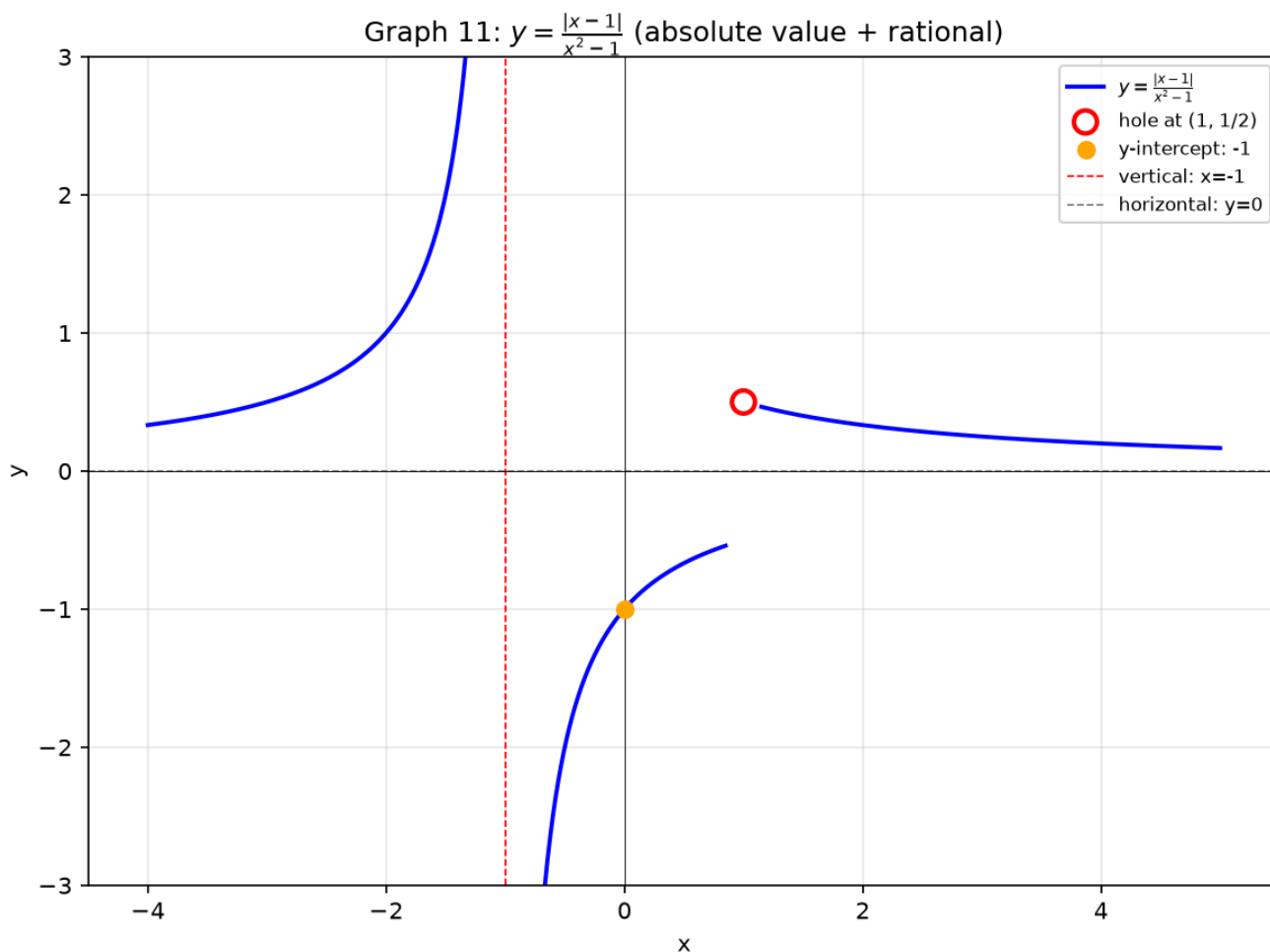
Asymptotes: vertical $x = -1$, horizontal $y = 0$ (both sides).

Drawing:

$x < -1$: $y = -\frac{1}{x+1}$. $x \rightarrow -1^- \rightarrow +\infty$, $x \rightarrow -\infty \rightarrow 0$.

$-1 < x < 1$: $y = -\frac{1}{x+1}$. $x \rightarrow -1^+ \rightarrow -\infty$, at $x = 0 \rightarrow -1$.

$x \geq 1$: $y = \frac{1}{x+1}$. Hole at $(1, \frac{1}{2})$, $x \rightarrow \infty \rightarrow 0$.



Common Mistakes

Mistake 1: Getting the Order Wrong for Inverse of a Composition

Wrong path: " $(f \circ g)^{-1} = f^{-1} \circ g^{-1}$ " $\leftarrow$ wrong.

Why it's wrong: The inverse of a composition reverses the order.

$$(f \circ g)^{-1} = g^{-1} \circ f^{-1}.$$

You put on socks then shoes. To take them off: socks first, shoes second.

Right path: Taking the inverse flips the inside-to-outside order.

Mistake 2: Forgetting to Punch a Hole After Canceling

Wrong path: " $\frac{(x-1)(x+2)}{x-1} = x + 2$, so it's just a line." $\leftarrow$ incomplete.

Why it's wrong: $x = 1$ made the original denominator 0 and can never go in.

You must not mark $(1, 3)$ on the canceled formula.

Right path: Punch an empty circle at $(1, 3)$.

Mistake 3: Filling the Right Endpoint of a Floor Function Step

Wrong path: "On $[1, 2]$, the value is 1." $\leftarrow$ wrong.

Why it's wrong: $[2, 0] = 2$. The interval where $[x] = 1$ is $1 \leq x < 2$.

$x = 2$ already belongs to the next step.

Right path: The right endpoint of the $[1, 2)$ interval is an empty circle.

Mistake 4: Believing a Graph Can Never Cross Its Asymptote

Wrong path: "There's a horizontal asymptote at $y = 1$, so the graph never touches $y = 1$." $\leftarrow$ not necessarily.

Why it's wrong: An asymptote only describes behavior as $x \rightarrow \pm\infty$.

For finite x , the graph can cross the asymptote.

Example: $y = \frac{x}{x^2+1}$ has horizontal asymptote $y = 0$ but passes right through the origin.

Right path: The graph only hugs the asymptote as $x \rightarrow \pm\infty$.

What We Just Did

- ① Find every x that can't go in. (square root, denominator, log, arcsin)
- ② When connecting two functions, compute from the inside out – push into the outer one.
- ③ For the inverse: solve $y=f(x)$ for x , then swap $x \leftrightarrow y$.
If one output comes from multiple inputs, snip the domain.
- ④ Draw graphs in 7 steps:
domain $\rightarrow$ symmetry $\rightarrow$ intercepts $\rightarrow$ asymptotes $\rightarrow$ sign $\rightarrow$ connect $\rightarrow$ special handling.
- ⑤ Transformations: move the base graph $(\pm h, \pm k)$,
flip $(-f, f(-x))$, fold $(|f|, f(|x|))$, stretch $(a \cdot f(bx))$.
- ⑥ $[x]$ makes stairs, $\{x\}$ makes sawteeth, piecewise functions are drawn interval by interval.

Exercise 1

Find the domain of $f(x) = \frac{\sqrt{9-x^2}}{\ln(x-1)}$.

Write the square root, log, and denominator conditions in order.

$\rightarrow$ Follow: **Example 2**

Solutions: [Solution Set](#)

Exercise 2

Draw the graph of $f(x) = \frac{x^2-4}{x^2-1}$ using the 7 steps.

Check asymptotes, intercepts, sign, and whether there's a hole.

$\rightarrow$ Follow: **Example 11**

Solutions: [Solution Set](#)

Exercise 3: Constructive

Make two functions of the form $\frac{ax+b}{cx+d}$ that both have asymptotes $x = 2$ and $y = 1$. Find their x - and y -intercepts, then compare and describe the two graphs.

→ Follow: **Example 13**

Solutions: [Solution Set](#)

Exercise 4: Constructive

Draw the graph of $f(x) = [x] + \{x\}^2$ on $[-2, 3]$. First write the formula for each interval $[n, n + 1)$.

→ Follow: **Examples 15, 16**

Solutions: [Solution Set](#)

Exercise 5

Draw the graph of $f(x) = \frac{|x|-1}{x-1}$. Split into $x \geq 0$ and $x < 0$ to strip the absolute value.

→ Follow: **Examples 20, 21, 24**

Solutions: [Solution Set](#)

Exercise 6: Challenge

Draw the graph of $f(x) = \frac{[x]}{x}$ on $[-3, 0)$ and $(0, 3]$ separately.

Use the fact that $[x]$ is constant on each interval to simplify the formula.

→ Follow: **Example 15**

Solutions: [Solution Set](#)

Today's Procedure

Step 1: Check the domain first.

Inside $\sqrt{} \geq 0$, denominator $\neq 0$, inside $\log > 0$, arcsin in $[-1,1]$.

If something cancels, record the hole location.

Step 2: Composition is inside-outside. Inverse: solve $y=f(x)$, swap $x \leftrightarrow y$.

If not one-to-one, snip the domain to make an inverse.

Step 3: Graph in 7 steps –

domain $\rightarrow$ symmetry $\rightarrow$ intercepts $\rightarrow$ asymptotes $\rightarrow$ sign $\rightarrow$ connect $\rightarrow$ special handling.

Transformations: move $(\pm h, \pm k)$, flip $(\pm f, f(\pm x))$,

fold $(|f|, f(|x|))$, stretch $(a \cdot f(bx))$.

Terminology

Up to now, we've only used simple words: "push in", "shove in", "move", "flip", "fold", "snip", "stick together".

You already know the methods. Now we give them their math names.

What we've been calling it	Math Term	Symbol / Explanation
numbers you can push in	domain	domain
numbers that come out	range	range
connecting functions	composite function	$(f \circ g)(x) = f(g(x))$

What we've been calling it	Math Term	Symbol / Explanation
swapping input and output	inverse function	$f^{-1}(x)$, symmetric across $y = x$
only one comes out per input	one-to-one (injective)	injective
origin symmetry	odd function	$f(-x) = -f(x)$
y -axis symmetry	even function	$f(-x) = f(x)$
stair function	floor function	$\lfloor x \rfloor = [x]$
ceiling function	ceiling function	$\lceil x \rceil$
sawtooth function	fractional part	$\{x\} = x - \lfloor x \rfloor$
sign function	signum function	$\text{sgn}(x)$
piece-by-piece function	piecewise function	different formulas on different intervals
moving the graph	translation	$f(x - h) + k$
flipping the graph	reflection	$-f(x), f(-x)$
folding upward	absolute value of function	$ f(x) $
stretching/shrinking	scaling	$a \cdot f(bx)$
hole	removable discontinuity	hole
line it hugs	asymptote	vertical, horizontal, slant